

Residue at Infinity

Brown & Churchill, Section 71

Setup

Let f be a meromorphic function such that all of its finite singularities lie inside a circle C . Then f is analytic outside C except possibly at ∞ , i.e. for

$$R_1 < |z| < \infty.$$

This is the setting for defining the *residue at infinity*.

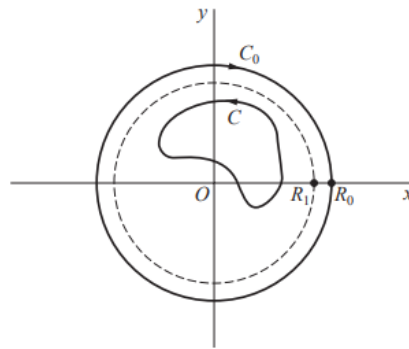


FIGURE 89

Orientation and the Residue Theorem

Let C_0 be a large circle enclosing all finite singularities of f , oriented clockwise.

By the residue theorem,

$$\oint_{C_0} f(z) dz = 2\pi i \operatorname{Res}(f, \infty)$$

$$\oint_C f(z) dz = 2\pi i \sum_{\text{finite } a_k} \operatorname{Res}(f, a_k).$$

Definition of the Residue at Infinity

Since

$$\oint_C f(z) dz = \oint_{-C_0} f(z) dz = - \oint_{C_0} f(z) dz.$$

$$\operatorname{Res}(f, \infty) = - \sum_{\text{finite } a_k} \operatorname{Res}(f, a_k)$$

Laurent Series Characterization

If f is analytic for sufficiently large $|z|$, it admits a Laurent expansion

$$f(z) = \sum_{n=-\infty}^{\infty} a_n z^n.$$

The coefficients are given by

$$a_n = \frac{1}{2\pi i} \oint_{-c_0} \frac{f(z)}{z^{n+1}} dz.$$

In particular,

$$a_{-1} = \frac{1}{2\pi i} \oint_{-c_0} f(z) dz.$$

Therefore,

$$\boxed{\text{Res}(f, \infty) = -a_{-1}}$$

That is, the residue at infinity is minus the coefficient of z^{-1} in the Laurent expansion of $f(z)$ at infinity.

Change of Variables Formula

Replace z by $1/z$ and multiply by $1/z^2$:

$$\frac{1}{z^2} f\left(\frac{1}{z}\right) = \sum_{n=-\infty}^{\infty} a_n z^{-n-2}.$$

The coefficient of z^{-1} in this series corresponds to a_{-1} . Hence,

$$\boxed{\text{Res}(f, \infty) = -\text{Res}\left(\frac{1}{z^2} f\left(\frac{1}{z}\right), 0\right)}.$$

Applied to problem 8.7.1d

$$\frac{\sin \frac{1}{z}}{z^2 + a^2}$$

Consider:

$$f(w) = \frac{1}{w^2} \frac{\sin w}{\frac{1}{w^2} + a^2} = \frac{1}{a^2} \frac{\sin w}{w^2 + 1/a^2}$$

Replace w by $1/z$ and multiply by $1/z^2$:

$$\frac{1}{z^2} f\left(\frac{1}{z}\right) = \frac{\sin \frac{1}{z}}{z^2 + a^2}$$

$$\text{Res}(f, \infty) = -\text{Res}\left(\frac{1}{z^2} f\left(\frac{1}{z}\right), 0\right) = -\text{Res}\left(\frac{\sin \frac{1}{z}}{z^2 + a^2}, 0\right)$$

Then:

$$\begin{aligned} \text{Res}\left(\frac{\sin \frac{1}{z}}{z^2 + a^2}, 0\right) &= -\text{Res}(f, \infty) = \sum_{\text{finite } a_k} \text{Res}(f, a_k) \\ &= \text{Res}\left(f, +\frac{i}{a}\right) + \text{Res}\left(f, -\frac{i}{a}\right) = \frac{1}{a^2} \frac{\sin\left(\frac{i}{a}\right)}{\frac{2i}{a}} + \frac{1}{a^2} \frac{\sin\left(-\frac{i}{a}\right)}{-\frac{2i}{a}} = \frac{\sin\left(\frac{i}{a}\right)}{ia} \end{aligned}$$

Or

$$\text{Res}\left(\frac{\sin \frac{1}{z}}{z^2 + a^2}, 0\right) = \frac{1}{a} \sinh\left(\frac{1}{a}\right)$$